Niranjan Purushothaman

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Professional Summary

I am a Computer Science graduate with experience in software development, application integration, backend development, database management and AI-powered solutions. I have Strong foundation in Software Development Life Cycle (SDLC), troubleshooting and technical documentation. Experienced in collaborating across teams to deliver reliable software solutions.

Education

Sathyabama Institute Of Science And Technology, Chennai, India **2022 – 2026**

Bachelor of Engineering (B.E.) in Computer Science and Engineering, ***CGPA: 8.05/10***

- Relevant Coursework: Database Management Systems, Software Engineering, Operating Systems, Computer Networks, Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Requirements Analysis.

Work Experience

Tectra Technologies, Chennai, India **Nov 2025 – May 2026**

Artificial Intelligence Intern (6 Months)

- Integrated Large Language Models (LLMs) into a CRM system to automate student interactions using a chatbot, OCR marksheet extraction tool, course recommendation system, and an assistant for maternal services offering doula and nanny booking facilities.
- Developed AI-powered voice and document intelligence pipelines using speech-to-text (WhisperX), NLP, LLMs, Python, and data extraction tools to convert unstructured audio data into structured CRM records for customer service.
- Built a full-fledged hospital chatbot for booking doctors and medical tests, answering health-related queries, and surfacing insights about hospital services.

Technical Skills

- **Programming Languages:** Python, Java, SQL
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
- **Application Development:** REST APIs, FastAPI, Node.js, React, Backend Development, Application Integration
- **Software Engineering:** SDLC, Requirements Analysis, System Integration, Application Architecture, Application Design, System Testing, Troubleshooting, Debugging, Technical Documentation, Change Management
- **Cloud & DevOps:** Git, GitHub, Docker, Kubernetes, Cloud Fundamentals, CI/CD Basics
- **Identity & Operations:** Role-Based Access Control (RBAC), Database Operations, Incident Management, Application Support, Version Control
- **AI Technologies:** API Integration, AI Agents, LLMs, Multi-Agentic Systems, RAG, WhisperX, Hugging Face, AI Frameworks, NLP, Computer Vision
- **Web Development:** HTML, CSS, JavaScript, React, Node.js, FastAPI, WebSockets, REST API
- **Databases:** MySQL, MongoDB, PostgreSQL, Vector Databases, Redis
- **Tools:** Git, Docker, Postman, VS Code, Kubernetes, Cursor IDE, Antigravity, Claude Code, Android Studio

Projects

AI Integration in CRM for Student Intelligence and Automation **2026**

- Enhanced a company CRM system by integrating AI-powered features, including conversational chatbots, voice intelligence, and automated data extraction, improving user experience and significantly reducing manual workload.
- Integrated AI features into the CRM using WhisperX, OCR, LLMs, and FastAPI to automatically extract student insights from call recordings and marksheets, and developed a context-aware conversational chatbot for automated student support and intelligent data retrieval.

AI-Based SDLC Assistant with Multi-Agent Architecture **2026**

- Developed an AI-powered assistant using a multi-agent architecture to automate the requirements gathering, planning, and system design phases of the Software Development Life Cycle (SDLC), reducing manual effort by 70%.
- Implemented a Manager Agent to coordinate and monitor the Requirement, Planning, and Design agents, validate generated outputs, detect inconsistencies, and trigger regeneration for accurate results.
- Built specialized AI agents using Ollama (LLM), LangChain, LangGraph, and FastAPI to analyze user inputs and automatically generate Software Requirements Specifications (SRS), project plans, and system design documentation.

RAG-Based Conversational Chat Assistant **2025**

- Developed a document-based conversational AI chatbot using LangChain, Azure OpenAI, DeepLake vector database, and Streamlit, enabling users to upload PDFs and interact with them using a Retrieval-Augmented Generation (RAG) framework.
- Implemented vector embeddings and semantic search to retrieve relevant document information and generate accurate, context-aware responses through an interactive web interface.

Research And Publications

AI-Based SDLC Assistant with Multi-Agent Architecture **2026**

IEEE International Conference on Communication, Computing and Electronics Systems (ICCCES)

- Published a research paper in an IEEE conference (indexed in IEEE Xplore) based on my final year project, presenting a multi-agent AI system for automating Software Development Life Cycle (SDLC) phases, demonstrating the application of large language models and intelligent agents to real-world software engineering automation.
- Designed specialized AI agents, including a Requirement Agent, Planning Agent, Design Agent, and Manager Agent, to collaboratively generate Software Requirement Specifications (SRS), project plans, and system architecture.

Publication Link

Certifications

- Introduction to Generative AI - Google Cloud
- Machine Learning - Dimensionality Reduction - IBM
- Fundamentals of Encryption And Quantum Safe Techniques - IBM

Hobbies and Carrier Interests

- Chess – Improves logical thinking and concentration.
- Strength Training (Gym) – Helps maintain fitness and discipline.
- Freelancing - Developed Multiple websites and projects for clients and students.
- Currently developing an application for a Problem statement that deals with Improper rent maintenance in rented rooms.

Languages

Tamil | English (Fluent) | Telugu